Software Engineering, Architecture & Operational Standards

Classification: Internal Corporate Directive · Version: 4.0 · Department: Engineering Tech

1. SDLC, Git Branching & Code Review Mandates

1.1 Branching Model: Acme Engineering operates on Trunk-Based Development. Feature branches must be short-lived (maximum lifespan of 48 hours) before being merged into the 'main' trunk.

1.2 Peer Review SLA: Every Pull Request (PR) requires a minimum of two (2) peer code reviews and approval from a designated Code Owner. PR reviews must be completed within 24 hours of submission.

1.3 Automated Test Coverage: Pull requests must achieve a minimum of 80% automated unit test code coverage. PRs that reduce overall repository test coverage are blocked automatically by CI pipelines.

2. API Design Guidelines & Semantic Versioning

2.1 Semantic Versioning: All internal and public microservice APIs must strictly follow Semantic Versioning (vMAJOR.MINOR.PATCH). Breaking schema changes require incrementing the MAJOR version.

2.2 Schema Definitions: All REST APIs must publish comprehensive OpenAPI 3.0 (Swagger) specifications. High-throughput inter-service communication must utilize gRPC with compiled Protocol Buffers (Protobuf).

2.3 Deprecation Policy: When an API endpoint is deprecated, it must be supported for a minimum of six (6) months with active deprecation headers (Sunset: date) prior to complete decommissioning.

3. Incident Severity Levels & On-Call Response SLAs

Engineering outages and software degradation are classified under four formal incident tiers:

3.1 SEV-1 (Critical Outage): Core service down, affecting >10% of global active users. Response SLA: On-call engineer must acknowledge within 15 minutes and join the incident bridge. Status updates published every 30 minutes.

3.2 SEV-2 (Major Degradation): Major feature impaired with no immediate workaround. Response SLA: 30 minutes acknowledgement, hourly status updates.

3.3 SEV-3 (Minor Defect): System degraded but workarounds exist. Response SLA: 4 hours.

3.4 Blameless Post-Mortems: For all SEV-1 and SEV-2 incidents, a blameless post-mortem document and root-cause analysis (RCA) must be published within 72 hours of incident resolution.

4. Deployment Windows, Canary Releases & Freezes

4.1 Allowed Deployment Windows: Production software releases are permitted Monday through Thursday between 9:00 AM and 3:00 PM EST.

4.2 Friday Deployments Banned: Production releases are strictly prohibited on Fridays after 12:00 PM EST and throughout weekends, except for urgent SEV-1 hotfixes approved by the Engineering VP.

4.3 Canary Rollouts: Microservice deployments must use progressive canary releases (5% traffic for 30 minutes, 25% for 1 hour, then 100% full rollout), automatically rolling back if error rates exceed 0.5%.

4.4 Holiday Deployment Freeze: A global change freeze takes effect annually from December 15 through January 3 to safeguard holiday commercial stability.

5. Observability, Logging & PII Masking Standards

5.1 Structured Logging: All applications must emit structured JSON logs with correlation IDs (`trace_id`, `span_id`) propagated across microservices using OpenTelemetry.

5.2 Strict PII Sanitization: Application logs must never contain sensitive customer PII, credit card numbers, passwords, or authentication bearer tokens. Automated regex scrubbing filters are enforced at the logging agent level.

5.3 Retention Windows: Production application logs are retained for 30 days in hot storage (Datadog) and archived for 365 days in cold cloud storage (AWS S3 Glacier) for regulatory compliance.